

On presentation of the human cerebral sulci from inside of the cerebrum.

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The human cerebral cortex is highly convoluted forming patterns of gyri separated by sulci. The cerebral sulci and gyri are fundamental in cortical anatomy as well as neuroimage processing and analysis. Narrow and deep cerebral sulci are not fully discernible either on the cortical or white matter surface. To cope with this limitation, I propose a new sulci presentation method that employs the inner cortical surface for sulci examination from the inside of the cerebrum. The method has four steps, construct the cortical surface, segment and label the sulci, dissect (open) the cortical surface, and explore the fully exposed sulci from the inside. The inside sulcal maps are created for the left and right lateral, left and right medial, and basal hemispheric surfaces with the sulci parcellated by color and labeled. These three-dimensional sulcal maps presented here are probably the first of this kind created. The proposed method demonstrates the full course and depths of sulci, including narrow, deep, and/or convoluted sulci, which has an educational value and facilitates their quantification. In particular, it provides a straightforward identification of sulcal pits which are valuable markers in studying neurologic disorders. It enhances the visibility of sulci variations by exposing branches, segments, and inter-sulcal continuity. The inside view also clearly demonstrates the sulcal wall skewness along with its variability and enables its assessment. Lastly, this method exposes the sulcal 3-hinges introduced here.

Cortical Morphological Networks Differ Between Gyri and Sulci.

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This study explored how the human cortical folding pattern composed of convex gyri and concave sulci affected single-subject morphological brain networks, which are becoming an important method for studying the human brain connectome. We found that gyri-gyri networks exhibited higher morphological similarity, lower small-world parameters, and lower long-term test-retest reliability than sulci-sulci networks for cortical thickness- and gyrification index-based networks, while opposite patterns were observed for fractal dimension-based networks. Further behavioral association analysis revealed that gyri-gyri networks and connections between gyral and sulcal regions significantly explained inter-individual variance in Cognition and Motor domains for fractal dimension- and sulcal depth-based networks. Finally, the clinical application showed that only sulci-sulci networks exhibited morphological similarity reductions in major depressive disorder for cortical thickness-, fractal dimension-, and gyrification index-based networks. Taken together, these findings provide novel insights into the constraint of the cortical folding pattern to the network organization of the human brain.

Brain development (sulci and gyri) as assessed by early postnatal MR imaging in preterm and term newborn infants.

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To investigate fetal brain development in vivo using early postnatal cranial MRI in term and preterm newborn infants. 51 infants, 1.5-T whole-body system, extremity coil, spin-echo images obtained in all three planes (T1- and T2-weighted). Independent review by two neuroradiologists (blinded for gestational age and medical history) regarding the development of 12 sulci and 10 gyri using a modified scoring

system. Gestational age was obtained either by ultrasound or from the first day of the last menstrual period. head circumference 90th percentile, cerebral malformation, chromosomal/metabolic disorder. Age at study ranged from 23 5/7-43 0/7 postmenstrual weeks. Brain maturation starts in the central area and proceeds towards the parieto-occipital cortex. The frontal cortex develops last. Transition to degrees 1, degrees 2 and, degrees 3 starts at week 25, 32 and 35 in the central cortex and is completed at week 34, 36 and 39, respectively. Our data compare favourably with the two previously published reports about brain maturation. Early MR imaging seems therefore to be suited to study maturation of the fetal brain. This may be of pathophysiological relevance in neonatal intensive care and neurological follow-up studies.

Gyrification from constrained cortical expansion.

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The exterior of the mammalian brain--the cerebral cortex--has a conserved layered structure whose thickness varies little across species. However, selection pressures over evolutionary time scales have led to cortices that have a large surface area to volume ratio in some organisms, with the result that the brain is strongly convoluted into sulci and gyri. Here we show that the gyrification can arise as a nonlinear consequence of a simple mechanical instability driven by tangential expansion of the gray matter constrained by the white matter. A physical mimic of the process using a layered swelling gel captures the essence of the mechanism, and numerical simulations of the brain treated as a soft solid lead to the formation of cusped sulci and smooth gyri similar to those in the brain. The resulting gyrification patterns are a function of relative cortical expansion and relative thickness (compared with brain size), and are consistent with observations of a wide range of brains, ranging from smooth to highly convoluted. Furthermore, this dependence on two simple geometric parameters that characterize the brain also allows us to qualitatively explain how variations in these parameters lead to anatomical anomalies in such situations as polymicrogyria, pachygyria, and lissencephalia.

The development of the sulci and gyri of the bovine cerebral hemispheres.

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This study contributes to the existing knowledge and includes a complete description of the gyri and sulci of the bovine cerebral hemisphere, the chronological order of appearance of these sulci and gyri and the ages at which the sulci and gyri reach their final pattern as seen in the bovine adult. The research was undertaken because this specific area has not been covered by anatomists in any depth. The results of this study will contribute to the more complete descriptions of the anatomy of the bovine adult, as well as of the embryological development of the animal. A number of bovine embryos and fetuses were collected from the abattoir. Serial sections of the embryos were studied histologically, while the brains of the fetuses were examined macroscopically or with a dissection-microscope. Up until day 58 the cerebral hemisphere is smooth. Then an indentation appears which marks the position of the Fissura sylvia. The Sulcus rhinalis lateralis appears at day 68. At day 90 the following grooves are visible: the Sulcus suprasylvius, the Sulcus cinguli, the Sulcus splenialis, the Sulcus genualis, and the Sulcus corporis callosi. At this age, all the gyri associated with these sulci develop, although the gyri are well developed only after the grooves have formed. At day 110 the Sulcus ectosylvius, the Sulcus marginalis, and the Sulcus presylvius are visible, once again with their associated gyri appearing a short time thereafter. At day 130

the following grooves are visible: the Sulcus coronalis, the Sulcus cruciatus, the Sulcus ansatus, and the Sulcus endomarginalis. The cerebrum of the foetus at this stage has well developed gyri. At day 160 the Sulcus proreus, the Sulcus ectomarginalis, the Sulcus suprasplénialis, the Sulcus endosplénialis, the Sulcus calcarinus, and the Sulcus endogenualis have developed. The final groove to form is the Sulcus rhinalis medialis. Then the brain looks like that of the adult animal.

Development of cerebral sulci and gyri in fetuses of cynomolgus monkeys (*Macaca fascicularis*).

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This study aimed to clarify the development of sulci and gyri on the external surface of the cerebrum of cynomolgus monkeys. Sulcus formation began with the appearance of the lateral fissure on embryonic day (ED) 70, followed by delineations of four cerebral lobes by the emergence of the parietooccipital sulcus, central sulcus, and preoccipital notch on EDs 80-90. The following primary sulci were then visible until ED 120: the superior temporal sulcus on ED 90; the intraparietal sulcus, lunate sulcus, inferior occipital sulcus, and arcuate sulcus on ED 100; and the principle sulcus on ED 110; the occipitotemporal sulcus, anterior middle temporal sulcus, and superior postcentral dimple on ED 120. These sulci demarcated the superior temporal gyrus on ED 90, the precentral gyrus, supramarginal gyrus, and angular gyrus on ED 100, and the inferior and middle temporal gyri, postcentral gyrus, superior parietal lobule, superior, middle and inferior frontal gyri, and inferior occipital gyrus on ED 120. Except for the intermediate and lateral orbitofrontal sulci, the sulci that appeared on ED 130 and thereafter were not related to the gyrus demarcations. Intriguingly, the brain markedly gained weight on EDs 100 and 120, corresponding to the embryonic ages when almost all gyri were visible. The results suggest that a rapid growth of the cerebrum involves convolutions of the gyri by a regular sequence of the sulcus formation in cynomolgus monkeys. This study further provides a standard of reference for normal development in the cerebral cortical morphology of cynomolgus monkeys.

Sulci and gyri are topological cerebral landmarks in individual subjects: a study of brain navigation during tumour resection.

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Surgical resection of brain tumours aims at the maximal safe resection of the pathological tissue with minimal functional impairment. To achieve this objective, reliable anatomical landmarks are indispensable to navigate into the brain. The neuronavigation system can provide information to target the location of the patient's lesion, but after the craniotomy, a brain shift and relaxation mismatch with it often occur. By contrast, sulci/gyri are topological cerebral landmarks in individual patients and do shift with the brain parenchyma during lesion removal, but remain independent from brain shift in relation to the sulci/gyri. Here, we present a case report of a novel strategy based on anatomical landmarks to guide intraoperative brain tumour resection, without using a standard neuronavigation system. A preoperative brain mapping of the peri-tumoural sulci by the MRI and surface reconstruction was followed by confirmation of the anatomical landmarks for the motor cortex using navigated transcranial magnetic stimulation. The resulting location was used as a seed for diffusion tensor imaging tractography to reconstruct the corticospinal tracts. These selected cortical landmarks (sulci/gyri) delimited the margins

of the two lesions and the specific location under which the corticospinal tract courses, thus facilitating monitoring of the peri-tumoural region during brain resection. In this case, 96% of the brain tumour from the pericentral somatomotor region was successfully removed without chronic post-operative motor impairments. This approach is based on cortical anatomy that is fixed during surgery and does not suffer from the brain shift that could misplace the lesion according to the neuronavigation system.

The macro-structural variability of the human neocortex.

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The human neocortex shows a considerable individual structural variability. While primary gyri and sulci are found in all normally developed brains and bear clear-cut gross structural descriptions, secondary structures are highly variable and not present in all brains. The blend of common and individual structures poses challenges when comparing structural and functional results from quantitative neuroimaging studies across individuals, and sets limits on the precision of location information much above the spatial resolution of current neuroimaging methods. This work aimed at quantifying structural variability on the neocortex, and at assessing the spatial relationship between regions common to all brains and their individual structural variants. Based on structural MRI data provided as the "900 Subjects Release" of the Human Connectome Project, a data-driven analytic approach was employed here from which the definition of seven cortical "communities" emerged. Apparently, these communities comprise common regions of structural features, while the individual variability is confined within a community. Similarities between the community structure and the state of the brain development at gestation week 32 lead suggest that communities are segregated early. Subdividing the neocortex into communities is suggested as anatomically more meaningful than the traditional lobar structure.

Morphometric analysis of the main brain sulci and clinical implications: Radiological and cadaveric study.

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Sulci and gyri of the cerebrum can be easily identified with the aid of radiology but are difficult to locate during surgical operations, owing to anatomical variations and the surgical approach of the sulci through a small aperture. Therefore, this study was performed to locate the main sulci of the brain by using various anatomical landmarks in cadaveric brain specimens and CT scan images. In 31 cadaveric brain specimens (17 right and 14 left hemispheres) from people of unknown sex, 21 parameters associated with important sulci of the brain were studied. CT scan images for 150 patients in three age groups were examined. The patient IDs were categorized into 50 patients in each of the following age groups: 20-40 yr, 41-60 yr and 61-80 yr. Ten parameters were studied. The data were statistically analyzed in SPSS software. In the cadaveric brain specimens, comparisons of right and left hemispheres indicated that only the posterior part of the calcarine sulcus showed a significant difference ($p = 0.0394$). In CT scans within each age group, comparison of the right and left sides in males and females showed significant differences for many parameters (e.g., calcarine sulcus to occipital pole: right $p = 0.0025$; left $p = 0.0009$). Comparisons between male and female parameters also showed significant differences. This study aids in identifying the important functional areas of the brain situated near the sulci, given that the sulci are connected to the gyral functions and act as a barrier for the gyri. The findings may facilitate

neurosurgery operations.

Cingulate sulcus morphology and paracingulate sulcus variations: Anatomical and radiological studies.

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The sulci and gyri found across the cerebrum differ in morphology between individuals. The cingulate sulcus is an important landmark for deciding the surgical approach for neighboring pathological lesions. Identifying the anatomical variations of anterior cingulate cortex morphology would help to determine the safe-entry route through neighboring lesions. In this study, magnetic resonance imaging data acquired from 149 healthy volunteers were investigated retrospectively for anatomical variations of the paracingulate sulcus. Also, human cadaveric brain hemispheres were investigated for cingulate and paracingulate sulcus anatomy. All participants had cingulate sulci in both hemispheres ($n = 149$, 100%). Three types of paracingulate sulcus patterns were identified: "prominent," "present," and "absent." Hemispheric comparisons indicated that the paracingulate sulcus is commonly "prominent" in the left hemisphere ($n = 48$, 32.21%) and more commonly "absent" in the right hemisphere ($n = 73$, 48.99%). Ten (6.71%) people had a prominent paracingulate sulcus in both the right and left hemispheres. Seven (4.70%) of them were male, and 3 (2.01%) of them were female. Paracingulate sulci were present in both hemispheres in 19 people (12.75%), of which 9 (6.04%) were male and 10 (6.71%) were female. There were 35 (23.49%) participants without paracingulate sulci in both hemispheres. Eleven (7.38%) were male and 24 (16.11%) were female. There were 73 (48.99%) participants without right paracingulate sulcus and 57 (38.26%) participants without left paracingulate sulcus ($p = 0.019$). In the examinations of the cadaver hemispheres, the paracingulate sulcus was present and prominent in 25%, and the intralimbic sulcus was present in 15%. It has been observed that the paracingulate sulcus is more prominent in the normal male brain compared to females. In females, there were more participants without paracingulate sulcus. This study shows that there are both hemispheric and sex differences in the anatomy of the paracingulate sulcus. Understanding the cingulate sulcus anatomy and considering the variations in the anterior cingulate cortex morphology during surgery will help surgeons to orient this elegant and complex area.
